Professional Report: Plant Disease Detection (PDT) Model Training Summary

This report summarizes the workflow, environment setup, dataset preparation, model training, validation,
and inference steps performed for building a Plant Disease Detection (PDT) system using YOLOv8.

1. Environment Setup:
A dedicated Python virtual environment was created inside:
C:/Users/Reddy/Documents/Minor Project/venv
PyTorch with CUDA 12.8 was installed and verified using GPU (NVIDIA RTX 3050, 6GB VRAM).
Ultralytics YOLOv8 was installed and validated.

2. Dataset Structure:
Dataset directory used:
C:/Users/Reddy/Documents/Minor Project/Datasets/PDT dataset/
Two subsets existed: LH and LL, each containing images and YOLO-format label files.
Healthy images generally contained empty label files; unhealthy images had bounding box annotations.

3. Label Verification:
A custom visualization script allowed inspection of YOLO bounding boxes on images to ensure label quality.
Single-image verification was done using OpenCV with normalized YOLO coordinates decoded into bounding boxes.

4. Model Training:
YOLOv8n (nano variant) was trained on the dataset using:
epochs=50, imgsz=640, batch=8, device=0 (GPU)
Training completed in ~4 hours.
Training saved to:
C:/Users/Reddy/Documents/Minor Project/runs/detect/train2/

5. Model Performance:
Best model saved as:
train2/weights/best.pt
Validation metrics:
Precision: ~0.879
Recall: ~0.861
mAP50: ~0.935
mAP50-95: ~0.662

6. Model Inference:
Inference on single test images was successfully performed using:
yolo task=detect mode=predict model=best.pt source= save=True show=True device=0
Predictions saved under:
runs/detect/predict/

7. GPU Verification:
Torch reported CUDA availability and allowed CUDA tensor allocations.
Ultralytics confirmed GPU acceleration for training and inference.

This concludes the PDT YOLOv8 training and evaluation workflow.